



- 9 (a) State what is meant by the binding energy of a nucleus.

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- (b) Table 9.1 shows the masses of two sub-atomic particles and a polonium-212 ($^{212}_{84}\text{Po}$) nucleus.

Table 9.1

	mass/u
proton	1.007 276
neutron	1.008 665
polonium-212 nucleus	211.942 749

For the polonium-212 nucleus, determine:

- (i) the mass defect Δm , in kg

$$\Delta m = \dots\dots\dots \text{ kg [3]}$$

- (ii) the binding energy

$$\text{binding energy} = \dots\dots\dots \text{ J [2]}$$

- (iii) the binding energy per nucleon.

$$\text{binding energy per nucleon} = \dots\dots\dots \text{ J [1]}$$





- (c) (i) On Fig. 9.1, sketch the variation with nucleon number A of binding energy per nucleon for values of A from 1 to 250.

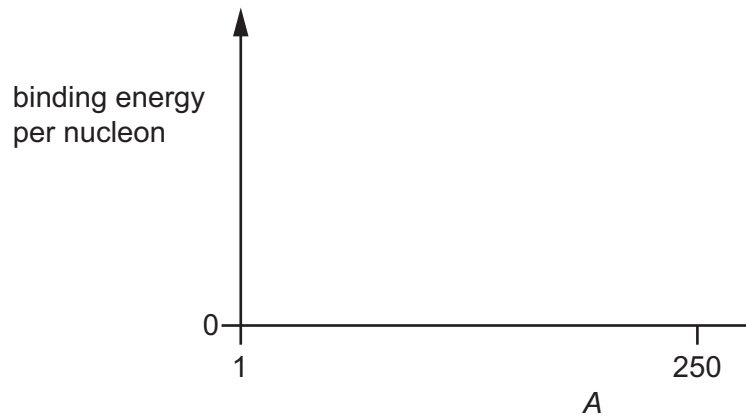


Fig. 9.1

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- (ii) On your line in Fig. 9.1, draw an X to show the approximate position of polonium-212.

[1]

- (iii) Polonium-212 is radioactive and undergoes alpha-decay.

Suggest and explain, with reference to Fig. 9.1, why the alpha-decay of polonium-212 results in a release of energy.

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[Total: 13]

